E-Commerce Funnel Analysis Using SQL & Power BI

EXECUTIVE SUMMARY

This project analyses user behaviour in an e-commerce website using funnel analysis. The goal is to understand how users move through different stages of the purchasing journey and identify where users drop off before completing a purchase.

The analysis was conducted using SQL for data processing and Power BI for visualization. The dataset contains user interaction events such as browsing products, adding items to the cart, checking out, and completing purchases.

The funnel analysis revealed the following key findings:

- A large number of users enter the funnel by browsing products.
- A noticeable drop occurs between the **Add to Cart** and **Checkout** stages.
- Only a small percentage of users complete the final **Purchase** stage.
- Some users leave the website early, contributing to the **bounce rate**.

Additional analysis was conducted to evaluate **device usage, marketing channels, and bounce behaviour**, helping to understand how different factors affect user engagement.

The insights from this analysis can help businesses:

- Improve the checkout experience
- Reduce drop-offs in the purchasing process
- Optimize marketing channels
- Improve website performance

Overall, this project demonstrates how **data analytics can help businesses understand customer behaviour and improve conversion rates through data-driven decision making**.

DATASET DESCRIPTION

The dataset used in this project is a synthetic dataset containing user interaction events on an e-commerce website.

Dataset Columns:

COLUMN	DESCRIPTION
User_ID	Unique ID of each user
Session_ID	Unique ID for each session
Event	User action in the funnel
Timestamp	Time when the event occurred
Device	Device used by the user
Region	User's Location
Channel	Marketing channel source
Product_Category	Category of the product
Revenue	Revenue generated from purchase
Bounce_Flag	Indicates if the user bounced

DATA LOADING AND PREPARATION

The dataset was first stored as a CSV file and then imported into a MySQL database.

Creating Table in MySQL

```sql
USE funnel_analysis;

CREATE TABLE funnel_data (
User_ID VARCHAR(20),
Session_ID VARCHAR(20),
Event VARCHAR(50),
Timestamp DATETIME,
Device VARCHAR(20),
Region VARCHAR(50),
Channel VARCHAR(50),
Product_Category VARCHAR(50),
Revenue FLOAT,
Bounce_Flag VARCHAR(10)
);
```

Loading the CSV File

```
LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/funnel_analysis_data.csv'
INTO TABLE funnel_data
FIELDS TERMINATED BY ','
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;
```

Data Cleaning

Before analysis, some cleaning steps were performed.

Fixing Timestamp Format

The timestamp was converted into MySQL datetime format.

Cleaning Bounce Flag

Some values had extra characters like yes\r and no\r.

```
UPDATE funnel_data
SET Bounce_Flag = TRIM(Bounce_Flag);
```

Converting Event Names to Camel Case

```
UPDATE funnel_data
SET Event =
CASE
WHEN Event = 'browse' THEN 'Browse'
WHEN Event = 'add to cart' THEN 'Add To Cart'
WHEN Event = 'checkout' THEN 'Checkout'
WHEN Event = 'purchase' THEN 'Purchase'
END;
```

FUNNEL STAGE ANALYSIS

```sql
SELECT Event, COUNT(*) AS Users
FROM funnel_data
GROUP BY Event
ORDER BY Users DESC;
```

This query counts how many users reached each stage of the funnel.

Event	Users
Browse	10000
AddToCart	7059
Checkout	3524
Purchase	1080

Insight

There is a significant drop in users between:

- Add to Cart → Checkout
- Checkout → Purchase

This indicates friction in the purchasing process.

CONVERSION ANALYSIS

```sql
SELECT
COUNT(DISTINCT CASE WHEN Event='Browse' THEN User_ID END) AS Browse,
COUNT(DISTINCT CASE WHEN Event='Add To Cart' THEN User_ID END) AS AddToCart,
COUNT(DISTINCT CASE WHEN Event='Checkout' THEN User_ID END) AS Checkout,
COUNT(DISTINCT CASE WHEN Event='Purchase' THEN User_ID END) AS Purchase
FROM funnel_data;
```

Browse	AddToCart	Checkout	Purchase
10000	7059	3524	1080

This query counts unique users at each funnel stage.

Insight

Only a small percentage of users who browse products actually complete the purchase.

This suggests opportunities to improve:

- **Checkout experience**
- **Payment process**
- **Product trust signals**

BOUNCE RATE ANALYSIS

Bounce rate measures how many sessions leave without interacting further.

```sql
SELECT
COUNT(*) AS Total_Sessions,
SUM(CASE WHEN Bounce_Flag = 'yes' THEN 1 ELSE 0 END) AS Bounce_Sessions,
(SUM(CASE WHEN Bounce_Flag = 'yes' THEN 1 ELSE 0 END) / COUNT(*)) * 100 AS Bounce_Rate
FROM funnel_data;
```

	Total_Sessions	Bounce_Sessions	Bounce_Rate
▶	21663	17343	80.0582

Insight

A high bounce rate suggests:

- **Poor landing page experience**
- **Slow website loading**
- **Irrelevant marketing traffic**

Reducing bounce rate can significantly improve conversions.

DEVICE ANALYSIS

```sql
SELECT Device, COUNT(*) AS Total_Events
FROM funnel_data
GROUP BY Device
ORDER BY Total_Events DESC;
```

This query shows which device users mostly use.

	Device	Total_Events
▶	Tablet	7237
	Desktop	7235
	Mobile	7191

Insight

Businesses can optimize their website for the most used device type.

For example:

- **If mobile traffic is high → improve mobile UX.**

MARKETING CHANNEL ANALYSIS

```
SELECT Channel, COUNT(*) AS Total_Events
FROM funnel_data
GROUP BY Channel
ORDER BY Total_Events DESC;
```

This query shows which channels bring the most users.

Channel	Total_Events
Google Ads	5435
Email	5433
Social Media	5412
Organic	5383

Insight

This helps businesses understand which marketing channels perform best, such as:

- **Organic search**
- **Social media**
- **Paid ads**
- **Email marketing**

Companies can allocate more budget to high-performing channels.

POWER BI DASHBOARD

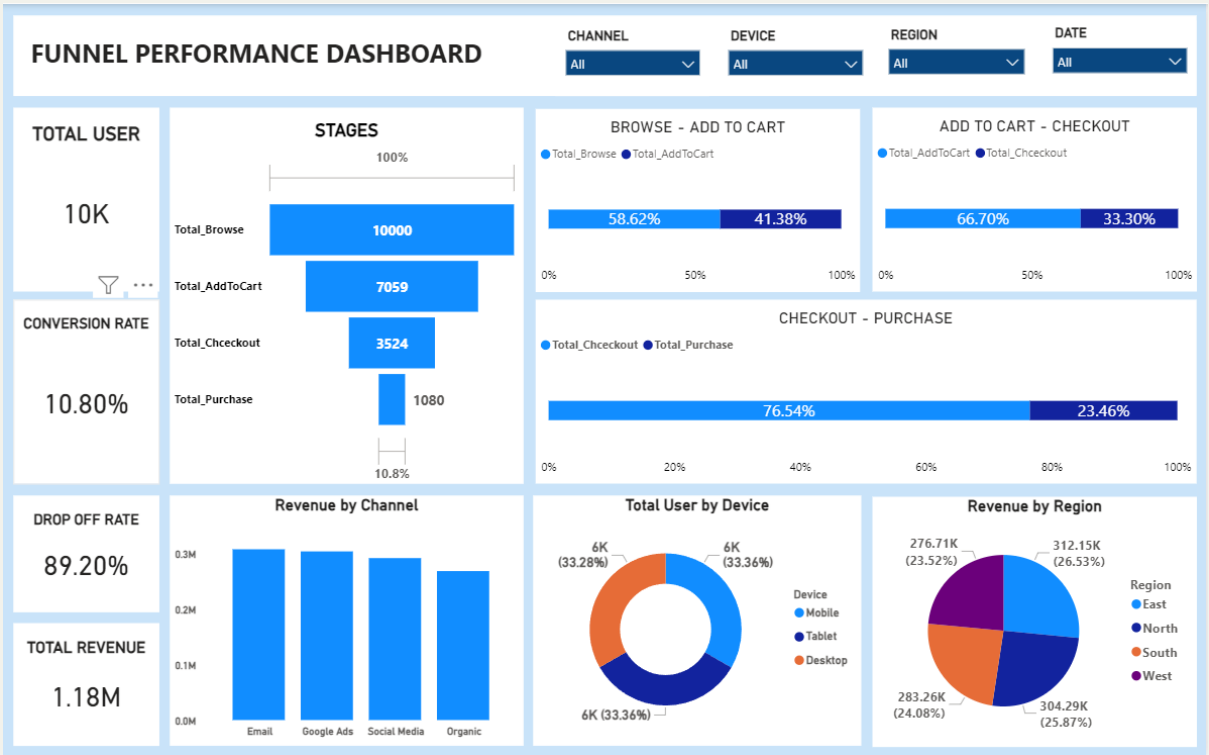
A Power BI dashboard was created to visualize key metrics such as:

- Funnel conversion stages
- Bounce rate
- Device distribution
- Marketing channel performance
- Revenue analysis
- Conversion Rate

Visualizations used:

- Funnel chart
- Bar charts
- cards
- Pie charts
- Sliders

These visuals help stakeholders easily understand user behaviour.



KEY INSIGHTS

1. **A large number of users drop off after the Add to Cart stage.**
2. **The Checkout to Purchase conversion is relatively low.**
3. **Some users leave the website immediately, contributing to the bounce rate.**
4. **Certain marketing channels bring more traffic than others.**
5. **Device type influences user behaviour and conversion patterns.**

BUSINESS RECOMMENDATIONS

Based on the analysis, businesses can improve conversions by:

- **Simplifying the checkout process**
- **Improving page loading speed**
- **Optimizing mobile experience**
- **Offering better discounts or incentives**
- **Improving trust signals such as reviews and secure payment options.**

CONCLUSION

This project demonstrates how SQL and Power BI can be used to analyse user behaviour in an e-commerce funnel.

By analysing funnel stages, bounce rate, and marketing channels, businesses can identify problem areas and improve their conversion rates.

Funnel analysis is an important tool for data-driven decision making in digital businesses.